

Evolution in acute pain assessment and treatment in the paediatric emergency department of a tertiary health care centre

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Abstract

Background: Paediatric pain remains one of the most misunderstood, under-diagnosed and under-treated medical problems in children.

Aim: To investigate the accuracy of acute pain assessment and management in the Paediatric Emergency Department in Lithuanian University of Health Sciences Hospital.

Methods: We performed a retrospective record analysis before (the year 2017) and after (the year 2018) paediatric pain training course was conducted. In total, 1,000 randomly selected outpatient records were analysed. We divided all patients into two groups: group A records from 2017 and group B from 2018. Patients were further divided into trauma and non-trauma and subdivided into four different age groups. We collected patient age, the origin of pain, pain characteristics, pain score and medication.

Results: We compared 500 children in each group. Groups A and B consisted of 154 (30.8%) and 116 (23.2%) trauma patients, respectively. The pain was scored less in group A (420 children; 84%) compared to group B (94.4% of all 500 patients, $p < .001$). In all age groups, the pain was assessed more often, and pain medication was prescribed more often in group B compared to group A ($p < .001$). There was a tendency to assess pain more often in group A non-trauma patients ($p = .054$). However, pain relief in trauma patients was less adequate compared to non-trauma.

Conclusion: Our research showed improvement in pain evaluation and treatment after systemic and local changes in paediatric emergency department (PED). In group B, pain was evaluated more frequently, and patients received pain medication more often than in group A. Teenagers are still less likely to receive analgesics than toddlers. Tendency remains to give fewer painkillers to trauma patients compared to non-trauma children.

Significance: This is one of the few studies in the Baltic region analysing changes in children's pain assessment and management in the PED after systemic and local adjustments. Our data show an impact of systemic changes and a specialized education focused on improving knowledge in acute pain diagnostics and management of medical professionals leading to better patient care.

1 | INTRODUCTION

In the past, the International Association for the Study of Pain defined pain as “an unpleasant sensory and emotional experience associated with actual or potential tissue damage or described in terms of such damage” (Loeser and Treede, 2008). But, the modern definition of pain has changed completely. Today, pain is characterized not only as physiological reaction and response to tissue damage but also as a complex phenomenon of physiological and pathological alterations in the nervous system. Paediatric pain is recognized as a multidisciplinary problem (Skausmomedicina.lt, 2018). Insufficient pain relief may cause long-term changes in pain understanding and perception, and lead to specific pain-related behavioural expressions (Srouji, Ratnapalan, & Schneeweiss, 2010). It has been shown that patients with recurrent abdominal pain in childhood were more likely to endorse anxiety symptoms and disorders, hypochondriacal behaviour, meagre social functioning and psychoactive medication overuse in adulthood (Campo, Di Lorenzo, & Chiappetta, 2001). Thus, there is an urgent need to improve children's pain management concerning human ethical and moral principles considering that pain affects not only children's health but may potentially contribute to their disability and suffering later in life (Finley, Franck, Grunau, & von Baeyer, 2005).

Pain is one of the most common reasons why children are referred to paediatric emergency department (PED) (Dale & Bjørnsen, 2015). Assessment of acute paediatric pain is challenging, especially in the crowded PED environment (Ali et al., 2014). Variations in patient's age and gender, development, and communication level, different personalities and temperaments, individual clinical condition and his/her personal response to a painful stimulus makes pain evaluation more complex. Moreover, past experiences related to primary care and/or hospital settings must be accounted for as well (Srouji et al., 2010). Education and skills of medical staff on pain identification and management can vary from PED to PED. Inadequate knowledge or skills during medical training or continuous education, personal bias and attitude towards pain (e.g. “pain medication will mask clinical condition”), underuse of pain-scoring tools or failure to recognize the need of medication, lack of pain reduction protocols as well as peculiarities of a local healthcare system can alone or all together be as the obstacles to optimal pain control (Schechter, 1989; Walco, Cassidy, & Schechter, 1994). Insufficient pain relief is more common in trauma patients compared to non-trauma (Saunders, Adelgais, & Nelson, 2010). Joseph and Brill (1999) reported that 60% of adults and only 28% of children presenting with long bone fractures received adequate analgesia in the emergency department (ED). Alexander et al. discovered that children younger than 2 years of age received disproportionately less analgesia than school-age children, despite having painful conditions (Alexander & Manno, 2003). In this study, pain management was compared

between very young children (up to 2 years) and school-age group. According to the research, pain killers were prescribed more often for the school-age children than for younger ones for all injuries (64.6% vs. 47.6%, respectively), all fractures (70.6% vs. 48.8%, respectively), displaced fractures (55.0% vs. 22.0%, respectively) and all burns (50.0% vs. 25.0%, respectively) (Alexander & Manno, 2003). Different data show that paediatric pain control remains suboptimal worldwide.

Research analysing acute paediatric pain recognition, management and/or care in Lithuanian hospitals and primary care institutions was still lacking. Moreover, less was known about specific medical personnel training programmes in pain medicine in the Baltics. Since 2013, systemic changes in pain management and care have been initiated on a national level in Lithuania. Several improvements have been made as follows. First, the major revision of nurses' professional expertise and skills concerning ED and PED was completed. Furthermore, an acute trauma patient care implementing trauma team protocol was approved. And finally, a new trauma centre and adult and paediatric emergency care centre has started to function in Lithuanian University of Health Sciences Hospital Kauno Klinikos (LSMU KK). Nevertheless, since the end of 2017, a pain-oriented medical staff education programme was started in LSMU KK paediatric emergency. Our study aimed to investigate the effect on acute pain assessment and analgesia in our PED before and after a specialized paediatric training programme was initiated in LSMU KK.

2 | METHODS

2.1 | Study design

The two-phase retrospective study was conducted in the PED of LSMU KK, Lithuania. The first phase of the study was performed in early 2017 following the initial pain and trauma care changes on national and local (tertiary hospital) level (“First phase of the study”). After the first results were obtained and analysed, local (PED) changes in paediatric pain recognition and management were initiated (“Second phase of the study”). Since these interventions were achieved, the second phase of our research started in 2018. Permission to conduct this study was issued by Kaunas Regional Biomedical Research Ethics Committee (Protocol No.: BE-2-54) and the study was conducted in accordance with the Declaration of Helsinki and good clinical practice guidelines.

2.2 | Study population

All paediatric patients who referred to PED in early 2017 (further referred to as “group A”) and the same period in

2018 (further were referred to as “group B”) were randomly selected for the study. Patients who had chronic diseases required hospitalization or refused treatment were excluded. The remaining patients were further divided into trauma and non-trauma patients (Figure 1). Also, all patients were subdivided according to their age as follows: up to 2 years, babies; 2–4 years, toddlers; 5–10 years, grade-schoolers and 11–17 years, teenagers (American Academy of Pediatrics, 2019). Demographic data, such as patient's age, and main complaint on arrival were obtained from the medical records.

2.3 | Pain evaluation

The pain was evaluated with different pain scales used in our PED as defined in our local PED protocol. Pain scales according to the child's age were used as following: FLACC (face, legs, activity, cry, consolability), Wong-Baker faces pain rating scale (Wong-Baker FACES), VAS (visual analogue scale) and VRS (verbal rating scale) (Kévalas, 2015). Initial data about pain assessment and management in PED were obtained from the patient medical records. The reason for pain (trauma/non-trauma), pain score, adequacy of pain relief according to the pain type, dosage of pain medication and follow-up data were collected.

2.4 | The first phase of the study

The first phase of the study was initiated in early 2017 after the systemic changes in emergency and trauma care were made on the national level in Lithuania. Trauma care and emergency care methodology was approved already after July 2014. But the implementation process extended until 2016. November 2016 the regulation regarding acute trauma care was achieved in our tertiary centre resulting in trauma team (adult and paediatric) formation and follow-up. The consensus was met that in PED a paediatrician as a leading physician in pain evaluation together with paediatric nurses, paediatric surgeon and traumatology fellow/doctor were accounted to manage acute moderate trauma patients. And a paediatric surgeon or a paediatric intensive care specialist together with traumatologist and paediatric nurses were in charge of the patients after severe and life-threatening events/trauma. Moreover, the first emergency medicine fellows started to work in PED from 2016. Besides, sixteen 24-hr follow-up beds were integrated into the new PED facility. We forecasted that previously mentioned changes would lead to improvement of pain and general condition management in patients with moderate and severe trauma. However, patients with mild trauma were still consulted by a traumatology fellow. Furthermore, data were lacking about pain diagnostics and management of children other than trauma referring to PED with pain as a chief complaint.

2.5 | Pain education programme and the second phase of the study

After the first data analysis and results of phase one, we made a decision to improve pain assessment and treatment. The pain evaluation and management programme were initiated consisting of the following steps: medical staff education and training, optimization of protocols on procedural sedation and acute pain evaluation and treatment. Herewith, pain research was initiated. Additionally, triage nurses together with a PED doctor/fellow rather than traumatology fellow were appointed to evaluate patients with mild trauma. In parallel, the electronic medical record system (e-MED) was implemented and started to function in our PED.

The education and training sessions for triage and procedural nurses, paediatric emergency physicians and fellows were established since early 2017 (after the first phase of the study). The up-to-date literature on paediatric pain scoring and analgesia (including both pharmacological and non-pharmacological pain relief) was presented and discussed monthly. Moreover, preceding critical patients with a possibility of feedback from paediatric traumatology/surgery/intensive care departments were discussed during PED morning rounds.

Following the implementation of e-MED, pain evaluation and documentation became a duty of the triage nurses. Also, nurses were assigned for a pain re-evaluation and documentation every 2 hr at the follow-up ward. The physician/fellow in charge was informed after initial pain evaluation, and proper medication was prescribed.

Protocols on procedural sedation, acute pain evaluation and treatment, and follow-up were reviewed, discussed and optimized (Hauer & Jones, 2018; Kévalas, 2015). After all of implementations in our department, it was decided locally that the target VAS score should be less than 4. The main reason of this decision was based on possibility to use non-pharmacological pain treatment in mild pain cases (Tables 1 and 2).

2.6 | Endpoints

The primary endpoint was to evaluate the difference if any in pain assessment and management in trauma and non-trauma patients considering different age (see “Study population”) before and after implementation of the local pain management programme. The second endpoint was to compare documentation and pain evaluation in the follow-up wards.

2.7 | Statistical analysis

Statistical analysis was performed using Microsoft Excel and IBM SPSS Statistics version 21.0 software (SPSS Inc.)

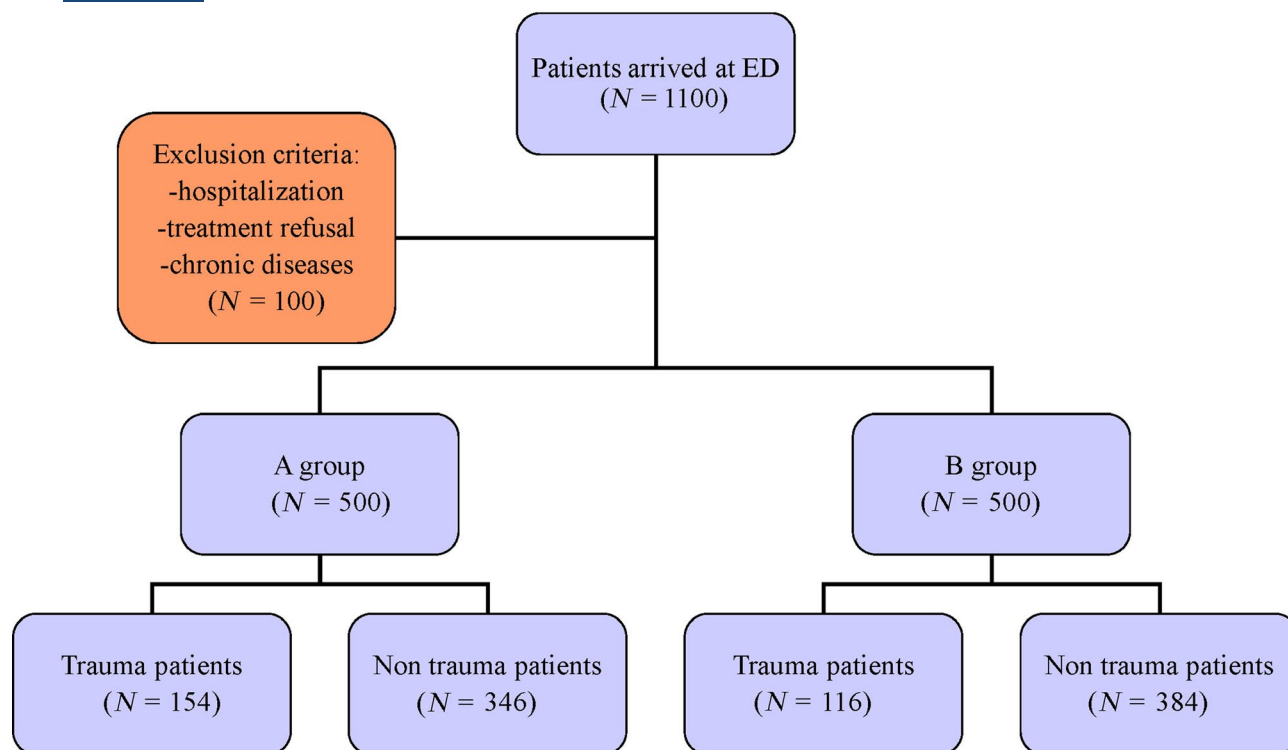


FIGURE 1 Methods of the study. Exclusion criteria and dividing into the groups. A digram reporting flow of retrospective case records. N—number of the patients

TABLE 1 Basic usage of medication according to pain score (Narayan, 2010; Stang et al., 2014)

Pain intensity	Pain management
1–3	Non-pharmacological, lidocaine ointment
4–6	Acetaminophen and/or NSAID
7–10	Opioid

for Windows. Descriptive statistical methods were applied, Kolmogorov–Smirnov test, T criterion for calculating pairs of samples, Mann–Whitney U test and Chi-square test. A p value of $<.05$ was considered significant.

3 | RESULTS

3.1 | General findings

In total, 1,100 randomly selected medical records of patients who arrived in PED were collected. After the exclusion due to chronic diseases, hospitalization and treatment refusal, 1,000 children were submitted for further analysis. The medical records meeting our inclusion criteria were further subdivided into two groups as follows: 500 in group A and 500 in group B. There was no significant difference in children age

between A and B groups (6.6 ± 5.4 years vs. 6.3 ± 4.9 years). Time of a first examination by a PED physician was better documented in group B compared to group A (56.2% vs. 46.4%). Trauma was the main reason for referral to PED in group A (30.9%). In group B, 23.2% of children visited PED due to trauma. The most frequent cause of referral other than trauma in group B compared to group A were upper respiratory tract infections (44.2% vs. 28.3%). The other complaints were less common (see Figure 2).

3.2 | Assessment of pain intensity

One of the aims of our study was to check how frequently pain was measured in our PED and to compare the differences between the groups. A significant improvement has been observed comparing the assessment of pain in group A (2017) and after previously mentioned changes in pain care (group B). In group A, 84% of the patients received pain evaluation and it was significantly lower compared to group B, where the pain was scored in 94.4% of the children ($p < .001$). However, pain assessment between the patient groups regarding the diverse cause of arrival did not differ significantly ($p = .254$). Comparing pain evaluation between the age groups, pain was scored more often in group B. It was statistically significant in all age groups (95.5% vs. 76% for patients under 2 years of age, ($p < .001$); 94.2% versus.

TABLE 2 Usage of proper medications according to dosage pro kilogram and maximum daily dose (Narayan, 2010)

Medication	Administration and dosage		Maximum daily dose
Acetaminophen	Per os	10–15 mg/kg every 4–6 hr	75 mg/kg day ⁻¹
	Per rectum	10–20 mg/kg every 4–6 hr	75 mg/kg day ⁻¹
	i/v	Up to 2 years: 7.5–15 mg/kg every 6hr	60 mg/kg day ⁻¹
		>2 years: • <50 kg: 15 mg/kg dose ⁻¹ every 6 hr or 12.5 mg/kg dose ⁻¹ every 4 hr • ≥50 kg: 1,000 mg every 6 hr or 650 mg every 4 hr	75 mg/kg day ⁻¹ (not to exceed 3,750 mg/day) 4,000 mg/day
Ibuprofen	Per os	4–10 mg/kg every 6–8 hr	40 mg/kg day ⁻¹
Ketorolac	i/v	Up to 2 years: 0.5 mg/kg every 6–8 hr (not to exceed 48–72 hr of treatment) >2 years: 0.5 mg/kg as a single dose; maximum dose: 15 mg/dose; Note: A single dose of 1 mg/kg (maximum: 60 mg/dose) has also been suggested (APS, 2008).	
	i/m	>2 years: 1 mg/kg	30 mg/dose
Morphine	Per os	0.08 mg/kg dose ⁻¹ every 4–6 hr	
	i/m, i/v (preferred)	Initial: 0.05–0.1 mg/kg dose ⁻¹ ; usual frequency every 4–6 hr, titrate carefully to effect	0.1 mg/kg dose ⁻¹
	Continuous i/v infusion	Initial: 0.01 mg/kg hr ⁻¹ (10 mcg/kg hr ⁻¹); titrate carefully to effect	0.03 mg/kg hr ⁻¹ (30 mcg/kg hr ⁻¹)
Fentanyl	i/m, i/v	2–3 mcg/kg dose ⁻¹	An i/v should be in place with general anaesthesia, so the i/m route is rarely used but still maintained as an option in the manufacturer's labelling.
	Intranasal	≥10 kg: 1.5–3 mcg/kg	100 mcg/dose

Note: kg, kilogram; mcg, microgram; mg, milligrams; mg/kg, milligrams pro kilogram.

Abbreviations: i/m, intramuscularly; i/v, intravenously; Per os, orally; per rectum, rectally.

82.7% for patients aged 2- to 4-years old ($p < .001$); for patients aged 5- to 10-years old—95.5% versus. 90% ($p = .021$) and for patients aged 11–17 years—92.2% versus. 85.0% ($p = .029$)) (Figure 3). In group B, pain assessment was more common in both trauma and non-trauma patients (Figure 4).

3.3 | Pain management: administration and effect of pain relievers

The sufficient or insufficient effect of analgesia was estimated by a choice of proper medication and its dosage (pro kg) taking into account pain intensity and its dynamics after painkiller administration. We found that patients in group B received more adequate analgesia than in group A (85.0% vs. 69.8%, respectively) and it was statistically significant ($p < .001$). In group B, adequate dosage of pain treatment was given more

frequently in all age groups compared to group A (<2 years of age pain reliever was prescribed appropriately in 93.3% versus. 74% children [$p < .001$], at the age of 2–4 years 89.1% versus. 75.2% [$p < .001$], from 5 to 10 years—84.7% versus. 71% [$p < .001$], and at the age of 11–17 years—74.1% versus. 60% [$p = .002$]) (see Table 3). In group A, there was a significantly worse pain reduction in trauma compared to non-trauma patients ($p = .005$). Adequate analgesia was given to 88.8% ($n = 341$) of non-trauma pain and 72.4% ($n = 84$) due to trauma in group B. In group B, adequate pain reliever was prescribed more often for reasons other than trauma ($p < .001$).

3.4 | Pain re-evaluation

In group A, 54 patients were admitted to the follow-up ward; meanwhile, 32 children were in need of observation in group

B. Comparing pain assessment, patients were more often re-evaluated if admitted to observation ward in group B (87.5% of the children) compared to group A (50% of the children) ($p < .01$).

4 | DISCUSSION

This is the first research in the Baltic region showing the effect of combination measurements towards the improvement of children's pain assessment and management in PED. The increasing awareness of the prevalence and impact of

untreated pain in the paediatric settings and the growing body of evidence on this topic made us investigate how pain was treated in our PED. After a first phase of the study, we noticed that there were patients whose pain was unevaluated leading to no treatment.

Paediatric pain evaluation can be rather difficult and with regard to a specific national-cultural background can be even neglected due to misleading prejudice or beliefs. The first phase of our study revealed that pain was scored only in 84% of children. This led to a hypothesis that paediatric pain was still misunderstood and underrated in our setting. Moreover, comparing pain scoring according to age, worst pain assessment was in children up to 2 years of age and teenagers, with 76% and 85%, respectively. Our data revealed that medical staff lacked knowledge in proper pain recognition in children of younger age. Three main methods were used to assess pain intensity, such as self-report, observational and physiological. When assessing pain in young children, attention is paid to the child's behavioural aspects, such as crying, facial expressions, changes in body posture and movements. This evaluation method requires more knowledge from a healthcare professional, so, infants' and toddlers' pain is assessed worse than pain in older ones (Drendel & Brousseau, 2006). Drendel et al. surveyed 24,707 children visits to the ED with painful conditions in a period of 3 years and showed that only 44.5% of visits had documented pain scores with subsequent analysis. The younger age children were the most under-evaluated group (Drendel & Brousseau, 2006). Our data analysis in 2017 showed that 24% under 2 years old and 18% 2–4 years received no pain assessment at all. The ability of children to describe their pain increases with age, experience, and developmental changes. Self-report is believed to be the most accurate method to indicate the intensity of pain. Both verbal and non-verbal pain identification require a certain degree of cognition and language development so that a child can understand and provide reliable answers (Stang, Hartling, Fera, & Johnson, 2014). However, after further analysis, we observed that also teenager group of patients received worse pain scoring. According to our study results, in 2017, the pain was properly documented only in 85% of teenagers. We hypothesized that some cultural beliefs and prejudice of medical staff and parents/caregivers could this phenomenon. Some of the healthcare providers believe that female patients' pain can be highly manipulative resulting in less attention to female pain expressions (Narayan, 2010). After the implementation of local medical staff education programme and ongoing systemic changes, in 2018 we noted a 10% improvement in pain evaluation overall (94.4% of all medical records). In our opinion, this achievement was reached as the nurses were attributed to a leading role in pain assessment and re-evaluation. e-MED granted improved pain scoring as the pain scale (according

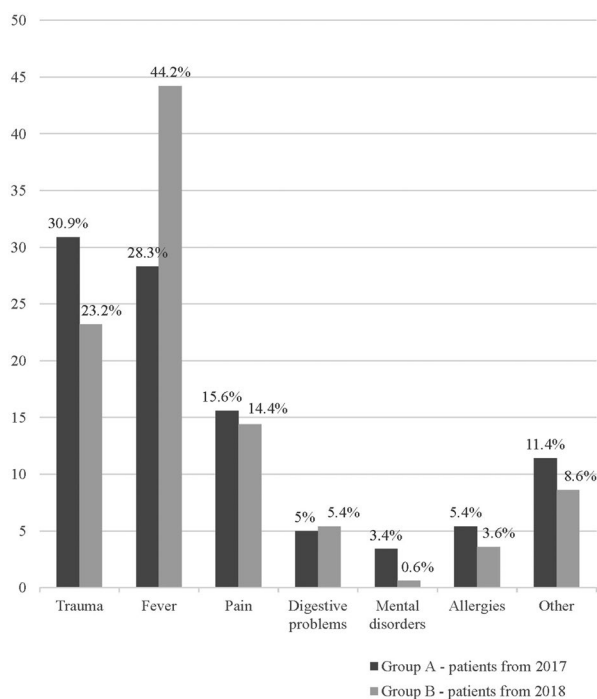


FIGURE 2 Causes of referral to the PED. PED, paediatric emergency department

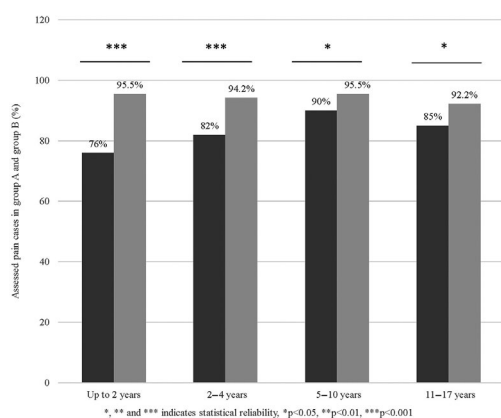


FIGURE 3 Pain assessment in the age groups of pediatric patients. * and *** indicates statistical reliability. * $p < .05$, *** $p < .001$

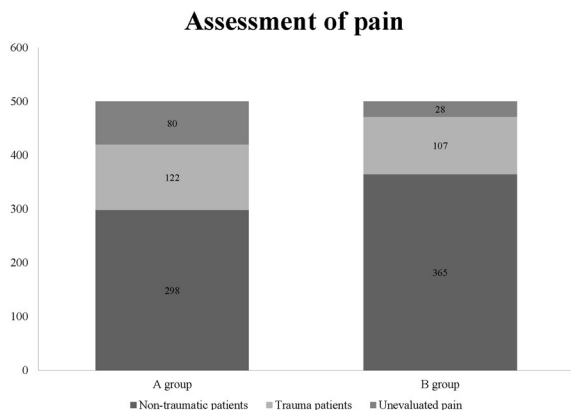


FIGURE 4 Assessment of different types of pain

TABLE 3 Adequate pain relief in patients of different age groups. All patients were subdivided according their age as following: up to 2 years, babies; 2–4 years, toddlers; 5–10 years, grade-schoolers and 11–17 years, teenagers

Age	Group A patients from 2017	Group B patients from 2018	p value
<2 years	74%	93.3%	<.001
2–4 years	75.2%	89.1%	<.001
5–10 years	71%	84.7%	<.001
11–17 years	60%	74.1%	.002

to age) and initial evaluation was made a routine procedure of triage nurses. As mild trauma patients were assigned to a PED physician/fellow, this led to better pain documentation and improved adequacy of pain medication. Still, around 6% of the patients received no pain scoring with a possibility that the pain which was not recognized and documented stayed under-treated or untreated. According to the research data, the main barriers to the proper pain scoring and treatment in children are as follows: (a) the myth that children, especially infants, do not feel pain the way adults do, or if they do, there is no untoward consequence; (b) lack of assessment and reassessment for the presence of pain; (c) misunderstanding of how to conceptualize and quantify a subjective experience; (d) lack of knowledge of pain treatment; (e) the notion that addressing pain in children takes too much time and effort and (f) fear of adverse effects of analgesic medications, including respiratory depression and addiction (American Pain Society, 2001). We do think that the main reason of those 6% patients whom were underscored could be a human factor related to a medical staff turnover, a short period to adapt to e-MED and improved pain scoring protocols. The overcrowding of our PED could contribute to additional problems first in adequate pain evaluation together with pain re-evaluation and

proper documentation (lack of time and personnel). Also, we noticed an improvement in teenagers' pain scoring up to 92.2% in 2018. However, pain assessment in this age group remained the worst compared to others. These data show that despite educational measurements and systemic changes some prejudice in teenager pain perception remained. We do think that it can be associated with changes within medical personnel, thus continuous educational programme regarding acute pain recognition and management should be repeated and introduced to newcomers. Moreover, parents/caregivers should be involved in paediatric pain education and be an active part of pain recognition as they are the best to know their children's personality and reaction to a painful stimulus. Insufficient recognition and evaluation of pain is a serious problem not only in our PED but also worldwide. A systematic review by Stang et al. noted that pain assessment ranged from 57% to 94% and only 23% of pain assessments used a validated pain scale. In this review, pain reassessment ranged from 32% to 50% (Stang et al., 2014). Another study made by Ferrante et al. showed that only 26% of Italian PED assess the intensity of pain. About one-third did not use algometric scales and almost half (47.4%) did not have local protocols for pain treatment (Ferrante et al., 2013). In our study, re-evaluation was performed in 50% of the patients in the year 2017 and increased to 87.5% in 2018. We suppose that such improvement was reached because of the training course and stricter rules of pain documentation. After the first phase of our study, nurses were assigned for pain re-evaluation and documentation every 2 hr at the follow-up ward.

Furthermore, we analysed the effectiveness of the pain-relief medication. Effective analgesia was determined when a proper medication and its dosage were prescribed according to pain intensity. The understanding and knowledge of proper pain reduction are still lacking especially considering paediatric trauma patients. Current pain management, however, is recognized as suboptimal (Kircher, Drendel, Newton, Dulai, & Vandermeer, 2014) and may be under-treated due to lack of clarity and competence as to which medication to prescribe (Ali et al., 2014; Ali, Williamson, Curtis, & Vandermeer, 2014). Lewis and Lasater (1994) retrospectively analysed 401 patients treated for acute bone fractures in eight PEDs. They concluded that only 30% of patients received proper analgesia. Jeremy et al. evaluated 208 children (aged 5 years) with burns referred to PED. Analgesia was prescribed only in 87% of the patients (Rawlins, Khan, Shenton, & Sharpe, 2007). According to our study, only 69.8% of patients in 2017 versus 85% in 2018 received adequate pain treatment. That correlates to worldwide recorded data (Kumle, Wilke, Koppert, & Kumle, 2013). Pain management in 2017 corresponded to assessment results and was the worst in infants, toddlers and teenagers (74% vs. 75.2% vs. 60%,

	Group A		Group B	
	<i>N</i>	% from total	<i>N</i>	% from total
All cases of under-diagnosed pain	80	16 (<i>n</i> = 500)	28	5.6 (<i>n</i> = 500)
Under-diagnosed pain cases among trauma patients	31	20.1 (<i>n</i> = 154)	9	7.7 (<i>n</i> = 116)
Under-diagnosed pain cases among non-trauma patients	49	14.2 (<i>n</i> = 346)	19	4.9 (<i>n</i> = 384)
All cases of under-treated pain	154	30.8 (<i>n</i> = 500)	75	15 (<i>n</i> = 500)
Under-treated pain cases among trauma patients	62	40.3 (<i>n</i> = 154)	32	27.6 (<i>n</i> = 116)
Under-treated pain cases among non-trauma patients	92	26.6 (<i>n</i> = 346)	43	11.2 (<i>n</i> = 384)

Note: Group A, patients from 2017; Group B, patients from 2018; *N*, *n* total number of patients.

TABLE 4 Appropriate pain evaluation and treatment between groups A and B

respectively). After training, a significant improvement in pain management was observed in infants reaching 93.3%. However, despite a 14.1% better pain relief in phase two, teenagers were the most under-treated group of children in 2018. As discussed previously, we think that due to a nationwide believes that older children can endure the pain without taking any medication, this group of children was lacking proper pain treatment. Moreover, we noticed that trauma pain remained under-treated compared to non-trauma patients both in A and B groups (Table 4). Still, a better pain assessment (a drop from 20.1% unevaluated trauma patients to 7.7%) and its management (from 40.3% under-treated trauma patients to 27.6%) was observed in 2018 versus 2017 including all age groups after the educational programme and systemic changes. A most significant change in trauma patient pain evaluation and treatment was trauma team implementation concerning severe trauma patients and pain scoring by triage nurses together with a PED doctor/fellow rather than traumatology fellow in mild trauma cases. However, the analgesia of trauma patients is still suboptimal. The injury itself, as well as its examination and treatment (i.e. radiographic tests and cast application), can cause significant pain and distress in a child. We speculate that a suboptimal assessment and treatment of trauma patients could be associated with fear of prescribing medication lacking knowledge in specific trauma patients (e.g. head-trauma or polytrauma patients).

5 | CONCLUSIONS

In conclusion, our study showed that systemic changes together with local measurements and medical staff training programme on pain assessment and treatment improved patient care. Still, examination time was not documented well

enough in both phases of the study. The initial pain evaluation and re-evaluation in the observation ward was performed more often in group B. Moreover, group B patients received pain relievers more often compared to group A and it was statistically significant. However, teenagers were less likely to be prescribed an adequate dosage of analgesic than younger children. Currently, pain intensity is more frequently assessed in non-trauma patients both in A and B groups. In both groups, pain reduction was more common in non-trauma patients and it was statistically significant when compared to trauma patients.

6 | RECOMMENDATIONS

1. To pursue better pain treatment. It is an urgent need to improve prescription of medication in infants and teenagers. All prejudice should be avoided by systemic training of healthcare professionals and education of patient's family members. Moreover, a patient's family should be involving in pain recognition and reduction process.
2. It is very important to have clear guidelines on pain assessment and treatment that are simple to use in crowded PED environment. Guidelines should be clear to all the PED team members including new ones. All information relayed on diagnostic or pain management should always be up to date.
3. Implementation of standard operating procedures of pain medication could lead to a better compliance of PED staff and make pain medication more consistent and efficient. It is especially important during the personnel turnover or when PED is overcrowded.
4. Accurate pain documentation is also one of the milestones in successful pain management. Well-documented pain improves collaboration between healthcare professionals

and reduces mistakes on the prescription of wrong medication.

5. It is also very important to re-evaluate pain after painkillers were administered even if the initial pain score was mild. Observation of pain dynamic matters for further pain control actions and patient's well-being.

7 | STRENGTHS AND LIMITATIONS

Our study is one of the few studies in the Baltic region showing the effect of combination measurements towards the improvement of children's pain assessment and management in PED. Our analysis was divided into two phases—before and after the before-mentioned changes. Even with a significant improvement, our study shows that this field needs further adjustment and observation. Recommendations concerning paediatric pain management could be prepared and implemented based on our study. The biggest limitation of our study is that this was a retrospective analysis in some patients' data were lacking what could lead to a result bias. Moreover, we could not measure if the proper pain scale was selected before e-MED was implemented (in phase one); so, we evaluated it according to our national and local recommendations. Pain evaluation was made using standard age-appropriate pain scales; however, pain scoring using them could be somehow subjective. One more limitation is that we did not divide trauma patients into groups by their trauma severity. Thus, we could not define the intensity of pain of different injuries. Also, results were not arranged gender-based, and we were unable to define any differences between female and male pain assessment and management. Gender and pain topics are still covered with a lot of prejudice which could influence pain evaluation and treatment.

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CONFLICTS OF INTEREST

The authors declare no conflict of interest.

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